

ics been around for ages — embedded in military and industrial equipment; refrigerators, washing machines and ovens too? Yes, but the splurge in applications and use-cases as we can see from the aforementioned examples is what has brought it back into focus. It is also important to understand that this clan of embedded electronics is quite different from the traditional applications that were implemented in industrial equipment.

Dr Arun Mulpur, manager-Industry Marketing, MathWorks Inc., explains, "There are several key differences between electronics in everyday objects and industrial machinery. The first is the sheer number of different applications and the environments and purposes for which electronics are being deployed in everyday functions. These include applications such as medical devices that require very high reliability and robustness as well as recreational applications such as devices embedded in footwear which may not need safety-critical embedded software. Industrial machinery applications—in contrast—are designed to operate in fairly controlled environments and perform repetitive tasks. Because of the safety and reliability considerations, these applications must adhere to very strict quality, efficiency, responsiveness, reliability and safety requirements."

Anand probes more into the technicalities, "From an engineering point of view, industrial machinery electronics performs a serious and critical role and has to adhere to stricter standards. It also needs to work 24x7 most of the times and has to be optimised for space, code size, power, etc. Additionally, it

is built to last for many years and is not cheap. On the other hand, daily electronics is typically (trend of these days) a use and throw type, comes at a cheaper price and does not need to be run 24x7. Moreover, it may not be optimised heavily due to the fact that daily electronics is more inclined towards application rather than critical parameters. Daily electronics is also becoming easily hackable by users and manufacturers are making it so happily; however, industrial machinery electronics still remains crack-proof, robust and proprietary in nature. Besides these, there are many other (major and minor) differences. However, the ones mentioned above are more distinctive."

"The level of qualification, certification and testing is much higher for industrial machinery as the duty cycle, environment and life cycle are very different," adds S. Natarajan, senior program manager-India Strategy & Planning, Intel.

Ugarkar adds, "There are two primary differences between embedded electronics in everyday objects and electronics embedded or used in industrial machinery. First, electronics

embedded in an everyday object must fuse or become one with the object. For example, electronics embedded in a jacket must not alter the shape or form of the jacket, so it not only needs to be small but also needs to be designed in such a way as to fuse or become one with the object. Second, design is extremely important. When I say design, I do not mean just the user interface aspects of the electronics but how well the addition of electronics continues to support the normal use of the objects as well as the additional features the electronics brings in." (See box 'Smart devices call for a smart design')

From this discussion, it is obvious that the current breed of embedded electronics, targeting everyday objects, is significantly different from the mission-critical embedded systems that have been common in the past. This opens up the need to understand hardware, software, networking protocols and design from a new perspective (See box 'What are the technical specifications of your saucepan?'). ●

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